Полный отчет анализа моделей

Сводные результаты

Серия	Модель	Средняя MAE	Средняя RMSE	Средняя sMAPE	Среднее время обучения	Лучшая модель
C	RNN (128u, 12w, 100e, 16b)	0.5752	1.2183	0.5660%	13.42 сек	
C	RNN (128u, 12w, 100e, 32b)	0.6330	1.2212	0.6233%	10.90 сек	
C	RNN (128u, 12w, 100e, 64b)	0.5293	1.2174	0.5204%	10.59 сек	
C	RNN (128u, 12w, 200e, 16b)	0.5611	1.2212	0.5520%	16.82 сек	
C	RNN (128u, 12w, 200e, 32b)	0.6041	1.2207	0.5947%	18.67 сек	
C	RNN (128u, 12w, 200e, 64b)	0.5338	1.2065	0.5249%	16.65 сек	
C	RNN (128u, 12w, 50e, 16b)	0.5508	1.2105	0.5418%	9.84 сек	
C	RNN (128u, 12w, 50e, 32b)	0.6305	1.2248	0.6208%	9.33 сек	
C	RNN (128u, 12w, 50e, 64b)	0.5325	1.2094	0.5236%	9.94 сек	
C	RNN (128u, 18w, 100e, 16b)	0.5849	1.2101	0.5756%	13.71 сек	
C	RNN (128u, 18w, 100e, 32b)	0.6027	1.2120	0.5933%	12.33 сек	
C	RNN (128u, 18w, 100e, 64b)	0.5649	1.2004	0.5557%	11.61 сек	
C	RNN (128u, 18w, 200e, 16b)	0.5712	1.2063	0.5620%	19.74 сек	
C	RNN (128u, 18w, 200e, 32b)	0.5696	1.2043	0.5604%	17.97 сек	
C	RNN (128u, 18w, 200e, 64b)	0.5687	1.2131	0.5595%	16.68 сек	
C	RNN (128u, 18w, 50e, 16b)	0.6254	1.2146	0.6158%	10.05 сек	
C	RNN (128u, 18w, 50e, 32b)	0.5214	1.1961	0.5126%	9.85 сек	
C	RNN (128u, 18w, 50e, 64b)	0.5423	1.1958	0.5333%	9.46 сек	
C	RNN (128u, 24w, 100e, 16b)	0.6296	1.2142	0.6198%	13.24 сек	
C	RNN (128u, 24w, 100e, 32b)	0.6098	1.2009	0.6001%	11.96 сек	
C	RNN (128u, 24w, 100e, 64b)	0.7925	1.2578	0.7810%	11.99 сек	
C	RNN (128u, 24w, 200e, 16b)	0.6075	1.1970	0.5978%	22.47 сек	
C	RNN (128u, 24w, 200e, 32b)	0.7217	1.2339	0.7110%	18.76 сек	
C	RNN (128u, 24w, 200e, 64b)	0.6911	1.2275	0.6807%	17.56 сек	
C	RNN (128u, 24w, 50e, 16b)	0.6526	1.2094	0.6426%	10.03 сек	
C	RNN (128u, 24w, 50e, 32b)	0.5912	1.1911	0.5817%	9.57 сек	
C	RNN (128u, 24w, 50e, 64b)	0.5894	1.1865	0.5799%	10.55 сек	
C	RNN (128u, 3w, 100e, 16b)	0.5714	1.2235	0.5622%	11.94 сек	
C	RNN (128u, 3w, 100e, 32b)	0.6795	1.2494	0.6694%	11.01 сек	

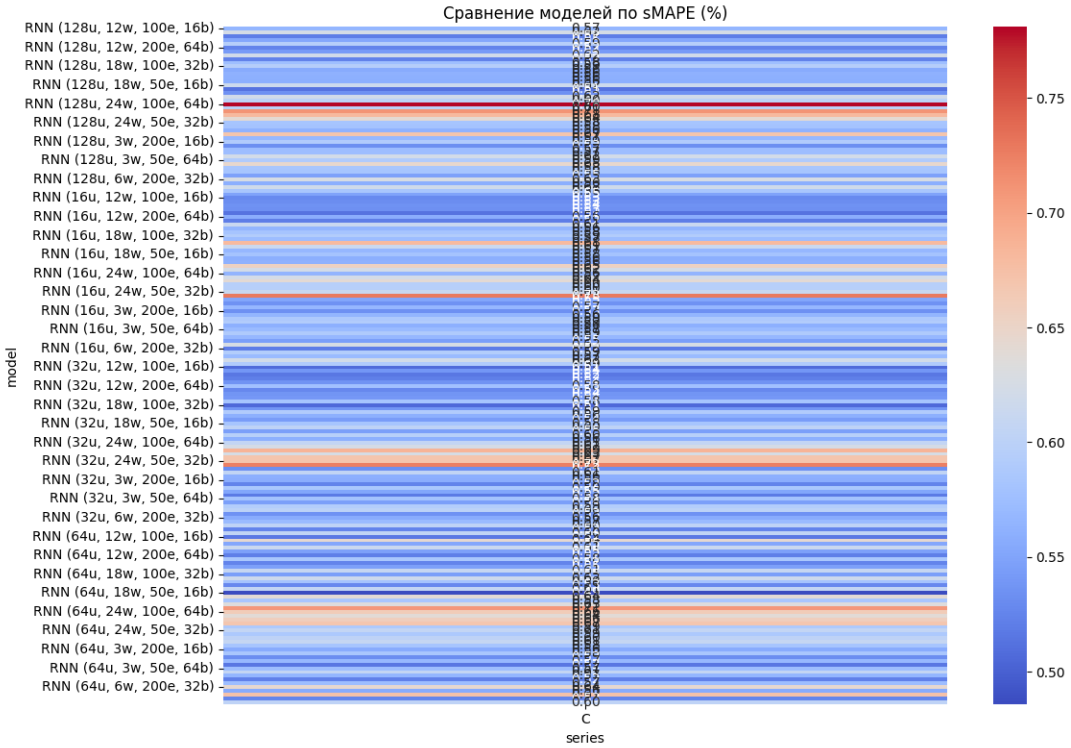
C	RNN (128u, 3w, 100e, 64b)	0.5770	1.2244	0.5678%	10.48 сек	
C	RNN (128u, 3w, 200e, 16b)	0.5997	1.2288	0.5903%	17.30 сек	
C	RNN (128u, 3w, 200e, 32b)	0.5418	1.2200	0.5328%	17.17 сек	
C	RNN (128u, 3w, 200e, 64b)	0.5826	1.2256	0.5734%	15.86 сек	
C	RNN (128u, 3w, 50e, 16b)	0.5823	1.2252	0.5730%	8.95 сек	
C	RNN (128u, 3w, 50e, 32b)	0.6307	1.2357	0.6211%	8.75 сек	
C	RNN (128u, 3w, 50e, 64b)	0.6000	1.2290	0.5906%	8.79 сек	
C	RNN (128u, 6w, 100e, 16b)	0.6549	1.2419	0.6451%	12.67 сек	
C	RNN (128u, 6w, 100e, 32b)	0.5930	1.2278	0.5837%	11.77 сек	
C	RNN (128u, 6w, 100e, 64b)	0.5919	1.2277	0.5826%	11.43 сек	
C	RNN (128u, 6w, 200e, 16b)	0.5564	1.2218	0.5474%	18.13 сек	
C	RNN (128u, 6w, 200e, 32b)	0.6374	1.2371	0.6277%	16.14 сек	
C	RNN (128u, 6w, 200e, 64b)	0.5683	1.2234	0.5591%	22.91 сек	
C	RNN (128u, 6w, 50e, 16b)	0.6291	1.2365	0.6195%	10.86 сек	
C	RNN (128u, 6w, 50e, 32b)	0.5927	1.2275	0.5833%	8.91 сек	
C	RNN (128u, 6w, 50e, 64b)	0.5573	1.2218	0.5482%	9.79 сек	
C	RNN (16u, 12w, 100e, 16b)	0.5387	1.2132	0.5297%	12.83 сек	
C	RNN (16u, 12w, 100e, 32b)	0.5406	1.2049	0.5317%	11.12 сек	
C	RNN (16u, 12w, 100e, 64b)	0.5444	1.2029	0.5354%	11.65 сек	
C	RNN (16u, 12w, 200e, 16b)	0.5429	1.2104	0.5340%	19.13 сек	
C	RNN (16u, 12w, 200e, 32b)	0.5218	1.2331	0.5130%	15.91 сек	
C	RNN (16u, 12w, 200e, 64b)	0.5670	1.1934	0.5579%	14.36 сек	
C	RNN (16u, 12w, 50e, 16b)	0.5292	1.1924	0.5203%	9.59 сек	
C	RNN (16u, 12w, 50e, 32b)	0.6218	1.2248	0.6122%	9.35 сек	
C	RNN (16u, 12w, 50e, 64b)	0.5740	1.2043	0.5647%	8.97 сек	
C	RNN (16u, 18w, 100e, 16b)	0.5900	1.1946	0.5807%	10.92 сек	
C	RNN (16u, 18w, 100e, 32b)	0.6011	1.2036	0.5917%	10.28 сек	
C	RNN (16u, 18w, 100e, 64b)	0.5819	1.1961	0.5726%	11.18 сек	
C	RNN (16u, 18w, 200e, 16b)	0.6909	1.2132	0.6806%	18.15 сек	
C	RNN (16u, 18w, 200e, 32b)	0.6227	1.2116	0.6131%	17.61 сек	
C	RNN (16u, 18w, 200e, 64b)	0.5764	1.2010	0.5672%	16.42 сек	
C	RNN (16u, 18w, 50e, 16b)	0.5896	1.2017	0.5803%	10.07 сек	
C	RNN (16u, 18w, 50e, 32b)	0.5682	1.2154	0.5591%	7.97 сек	
C	RNN (16u, 18w, 50e, 64b)	0.5710	1.1898	0.5617%	8.33 сек	
C	RNN (16u, 24w, 100e, 16b)	0.6647	1.2133	0.6546%	12.50 сек	

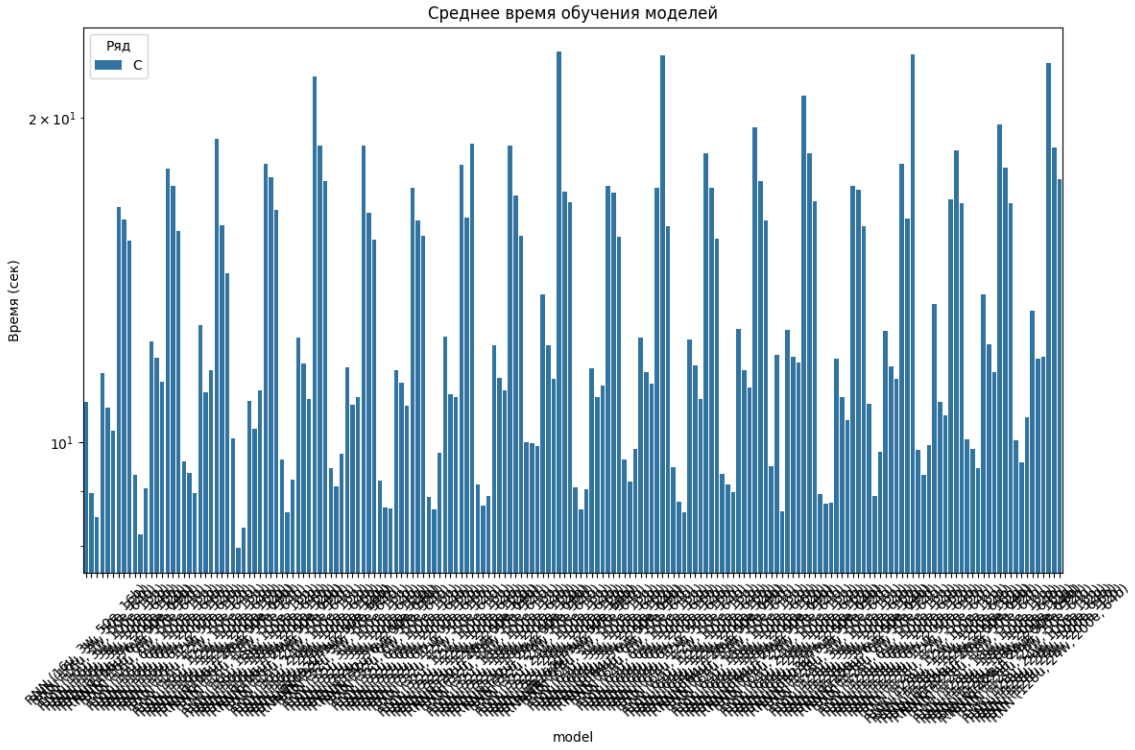
C	RNN (16u, 24w, 100e, 32b)	0.6315	1.2433	0.6217%	11.82 сек	
C	RNN (16u, 24w, 100e, 64b)	0.5726	1.1962	0.5632%	10.97 сек	
C	RNN (16u, 24w, 200e, 16b)	0.6323	1.2069	0.6226%	21.87 сек	
C	RNN (16u, 24w, 200e, 32b)	0.6510	1.2004	0.6410%	18.85 сек	
C	RNN (16u, 24w, 200e, 64b)	0.6057	1.1951	0.5961%	17.47 сек	
C	RNN (16u, 24w, 50e, 16b)	0.6058	1.2122	0.5963%	9.64 сек	
C	RNN (16u, 24w, 50e, 32b)	0.6211	1.2416	0.6114%	8.61 сек	
C	RNN (16u, 24w, 50e, 64b)	0.7395	1.2035	0.7291%	9.22 сек	
C	RNN (16u, 3w, 100e, 16b)	0.5600	1.2214	0.5509%	11.58 сек	
C	RNN (16u, 3w, 100e, 32b)	0.5438	1.2196	0.5348%	10.75 сек	
C	RNN (16u, 3w, 100e, 64b)	0.5763	1.2243	0.5671%	10.24 сек	
C	RNN (16u, 3w, 200e, 16b)	0.5428	1.2193	0.5339%	16.52 сек	
C	RNN (16u, 3w, 200e, 32b)	0.5736	1.2238	0.5644%	16.11 сек	
C	RNN (16u, 3w, 200e, 64b)	0.5975	1.2285	0.5881%	15.40 сек	
C	RNN (16u, 3w, 50e, 16b)	0.6005	1.2289	0.5911%	10.90 сек	
C	RNN (16u, 3w, 50e, 32b)	0.5698	1.2232	0.5606%	8.97 сек	
C	RNN (16u, 3w, 50e, 64b)	0.5800	1.2250	0.5708%	8.51 сек	
C	RNN (16u, 6w, 100e, 16b)	0.5984	1.2287	0.5890%	12.41 сек	
C	RNN (16u, 6w, 100e, 32b)	0.5762	1.2253	0.5671%	11.97 сек	
C	RNN (16u, 6w, 100e, 64b)	0.5544	1.2212	0.5454%	11.37 сек	
C	RNN (16u, 6w, 200e, 16b)	0.6358	1.2365	0.6261%	17.95 сек	
C	RNN (16u, 6w, 200e, 32b)	0.5292	1.2212	0.5204%	17.29 сек	
C	RNN (16u, 6w, 200e, 64b)	0.6015	1.2296	0.5921%	15.71 сек	
C	RNN (16u, 6w, 50e, 16b)	0.5798	1.2258	0.5706%	9.32 сек	
C	RNN (16u, 6w, 50e, 32b)	0.6307	1.2358	0.6211%	8.21 сек	
C	RNN (16u, 6w, 50e, 64b)	0.5997	1.2296	0.5903%	9.06 сек	
C	RNN (32u, 12w, 100e, 16b)	0.5163	1.2190	0.5075%	12.53 сек	
C	RNN (32u, 12w, 100e, 32b)	0.5473	1.2037	0.5383%	11.08 сек	
C	RNN (32u, 12w, 100e, 64b)	0.5239	1.2045	0.5151%	11.00 сек	
C	RNN (32u, 12w, 200e, 16b)	0.5295	1.2252	0.5207%	18.11 сек	
C	RNN (32u, 12w, 200e, 32b)	0.5449	1.2699	0.5359%	16.16 сек	
C	RNN (32u, 12w, 200e, 64b)	0.5855	1.2085	0.5763%	18.91 сек	
C	RNN (32u, 12w, 50e, 16b)	0.5372	1.2117	0.5283%	8.90 сек	
C	RNN (32u, 12w, 50e, 32b)	0.5462	1.2099	0.5372%	8.65 сек	
C	RNN (32u, 12w, 50e, 64b)	0.5481	1.2049	0.5391%	9.77 сек	

C	RNN (32u, 18w, 100e, 16b)	0.5916	1.2079	0.5822%	12.31 сек	
C	RNN (32u, 18w, 100e, 32b)	0.5144	1.1903	0.5056%	11.47 сек	
C	RNN (32u, 18w, 100e, 64b)	0.5470	1.1916	0.5380%	11.16 сек	
C	RNN (32u, 18w, 200e, 16b)	0.5996	1.1996	0.5902%	18.84 сек	
C	RNN (32u, 18w, 200e, 32b)	0.5693	1.1868	0.5601%	16.93 сек	
C	RNN (32u, 18w, 200e, 64b)	0.5524	1.1943	0.5433%	15.54 сек	
C	RNN (32u, 18w, 50e, 16b)	0.5918	1.2032	0.5825%	9.14 сек	
C	RNN (32u, 18w, 50e, 32b)	0.6107	1.2043	0.6012%	8.73 сек	
C	RNN (32u, 18w, 50e, 64b)	0.5549	1.1895	0.5459%	8.90 сек	
C	RNN (32u, 24w, 100e, 16b)	0.6051	1.1985	0.5955%	13.72 сек	
C	RNN (32u, 24w, 100e, 32b)	0.5696	1.1815	0.5602%	12.29 сек	
C	RNN (32u, 24w, 100e, 64b)	0.6158	1.1802	0.6062%	11.44 сек	
C	RNN (32u, 24w, 200e, 16b)	0.6341	1.2054	0.6242%	23.04 сек	
C	RNN (32u, 24w, 200e, 32b)	0.6969	1.2192	0.6865%	17.10 сек	
C	RNN (32u, 24w, 200e, 64b)	0.6358	1.1950	0.6260%	16.69 сек	
C	RNN (32u, 24w, 50e, 16b)	0.6805	1.2191	0.6702%	10.00 сек	
C	RNN (32u, 24w, 50e, 32b)	0.6806	1.2074	0.6704%	9.98 сек	
C	RNN (32u, 24w, 50e, 64b)	0.7356	1.2239	0.7248%	9.92 сек	
C	RNN (32u, 3w, 100e, 16b)	0.5414	1.2194	0.5325%	11.73 сек	
C	RNN (32u, 3w, 100e, 32b)	0.6148	1.2318	0.6053%	10.83 сек	
C	RNN (32u, 3w, 100e, 64b)	0.5681	1.2228	0.5590%	11.00 сек	
C	RNN (32u, 3w, 200e, 16b)	0.5674	1.2228	0.5583%	18.86 сек	
C	RNN (32u, 3w, 200e, 32b)	0.5336	1.2193	0.5247%	16.34 сек	
C	RNN (32u, 3w, 200e, 64b)	0.5951	1.2280	0.5857%	15.40 сек	
C	RNN (32u, 3w, 50e, 16b)	0.5597	1.2214	0.5506%	9.45 сек	
C	RNN (32u, 3w, 50e, 32b)	0.5262	1.2208	0.5174%	9.09 сек	
C	RNN (32u, 3w, 50e, 64b)	0.5894	1.2266	0.5801%	9.74 сек	
C	RNN (32u, 6w, 100e, 16b)	0.5538	1.2209	0.5448%	11.65 сек	
C	RNN (32u, 6w, 100e, 32b)	0.6017	1.2292	0.5923%	11.36 сек	
C	RNN (32u, 6w, 100e, 64b)	0.6096	1.2315	0.6002%	10.81 сек	
C	RNN (32u, 6w, 200e, 16b)	0.5436	1.2200	0.5347%	17.23 сек	
C	RNN (32u, 6w, 200e, 32b)	0.5656	1.2231	0.5565%	16.07 сек	
C	RNN (32u, 6w, 200e, 64b)	0.5799	1.2256	0.5707%	15.55 сек	
C	RNN (32u, 6w, 50e, 16b)	0.6123	1.2318	0.6028%	9.21 сек	
C	RNN (32u, 6w, 50e, 32b)	0.5334	1.2207	0.5245%	8.69 сек	

C	RNN (32u, 6w, 50e, 64b)	0.6137	1.2323	0.6042%	8.67 сек	
C	RNN (64u, 12w, 100e, 16b)	0.5258	1.2129	0.5169%	12.46 сек	
C	RNN (64u, 12w, 100e, 32b)	0.6529	1.2272	0.6430%	11.77 сек	
C	RNN (64u, 12w, 100e, 64b)	0.5614	1.2072	0.5523%	10.95 сек	
C	RNN (64u, 12w, 200e, 16b)	0.6227	1.2288	0.6131%	18.56 сек	
C	RNN (64u, 12w, 200e, 32b)	0.5553	1.2128	0.5463%	17.24 сек	
C	RNN (64u, 12w, 200e, 64b)	0.5304	1.2115	0.5215%	15.45 сек	
C	RNN (64u, 12w, 50e, 16b)	0.5935	1.2104	0.5842%	9.47 сек	
C	RNN (64u, 12w, 50e, 32b)	0.5331	1.2036	0.5243%	8.79 сек	
C	RNN (64u, 12w, 50e, 64b)	0.5550	1.2089	0.5460%	8.59 сек	
C	RNN (64u, 18w, 100e, 16b)	0.6190	1.2112	0.6094%	12.73 сек	
C	RNN (64u, 18w, 100e, 32b)	0.5549	1.2029	0.5458%	11.66 сек	
C	RNN (64u, 18w, 100e, 64b)	0.6264	1.2134	0.6167%	11.24 сек	
C	RNN (64u, 18w, 200e, 16b)	0.5898	1.2102	0.5804%	19.62 сек	
C	RNN (64u, 18w, 200e, 32b)	0.5365	1.1829	0.5275%	17.47 сек	
C	RNN (64u, 18w, 200e, 64b)	0.6163	1.2216	0.6067%	16.06 сек	
C	RNN (64u, 18w, 50e, 16b)	0.4946	1.1982	0.4860%	9.35 сек	+
C	RNN (64u, 18w, 50e, 32b)	0.6491	1.2245	0.6392%	9.12 сек	
C	RNN (64u, 18w, 50e, 64b)	0.5933	1.2029	0.5839%	8.97 сек	
C	RNN (64u, 24w, 100e, 16b)	0.6441	1.2006	0.6341%	12.72 сек	
C	RNN (64u, 24w, 100e, 32b)	0.7189	1.2168	0.7082%	12.00 сек	
C	RNN (64u, 24w, 100e, 64b)	0.6723	1.2098	0.6621%	11.86 сек	
C	RNN (64u, 24w, 200e, 16b)	0.6502	1.2025	0.6402%	20.99 сек	
C	RNN (64u, 24w, 200e, 32b)	0.6685	1.2165	0.6582%	18.55 сек	
C	RNN (64u, 24w, 200e, 64b)	0.6790	1.2079	0.6687%	16.73 сек	
C	RNN (64u, 24w, 50e, 16b)	0.5967	1.1921	0.5872%	9.50 сек	
C	RNN (64u, 24w, 50e, 32b)	0.6160	1.1919	0.6062%	12.04 сек	
C	RNN (64u, 24w, 50e, 64b)	0.5957	1.1971	0.5863%	8.61 сек	
C	RNN (64u, 3w, 100e, 16b)	0.6112	1.2311	0.6017%	11.70 сек	
C	RNN (64u, 3w, 100e, 32b)	0.6181	1.2326	0.6086%	11.01 сек	
C	RNN (64u, 3w, 100e, 64b)	0.5919	1.2274	0.5826%	11.27 сек	
C	RNN (64u, 3w, 200e, 16b)	0.5644	1.2223	0.5553%	17.29 сек	
C	RNN (64u, 3w, 200e, 32b)	0.5922	1.2275	0.5829%	17.05 сек	
C	RNN (64u, 3w, 200e, 64b)	0.5399	1.2194	0.5309%	15.50 сек	
C	RNN (64u, 3w, 50e, 16b)	0.5778	1.2245	0.5686%	9.08 сек	

C	RNN (64u, 3w, 50e, 32b)	0.5245	1.2240	0.5157%	8.65 сек	
C	RNN (64u, 3w, 50e, 64b)	0.5832	1.2255	0.5740%	9.03 сек	
C	RNN (64u, 6w, 100e, 16b)	0.6169	1.2324	0.6074%	12.49 сек	
C	RNN (64u, 6w, 100e, 32b)	0.5753	1.2247	0.5661%	11.62 сек	
C	RNN (64u, 6w, 100e, 64b)	0.5306	1.2208	0.5217%	11.33 сек	
C	RNN (64u, 6w, 200e, 16b)	0.5821	1.2258	0.5729%	17.23 сек	
C	RNN (64u, 6w, 200e, 32b)	0.6543	1.2422	0.6444%	22.89 сек	
C	RNN (64u, 6w, 200e, 64b)	0.5644	1.2229	0.5553%	15.88 сек	
C	RNN (64u, 6w, 50e, 16b)	0.6836	1.2500	0.6735%	9.62 сек	
C	RNN (64u, 6w, 50e, 32b)	0.5348	1.2200	0.5259%	9.19 сек	
C	RNN (64u, 6w, 50e, 64b)	0.6125	1.2318	0.6030%	9.85 сек	




Среднее время обучения моделей
Ряд
C
2×10^1
10^1
Время (сек)
model

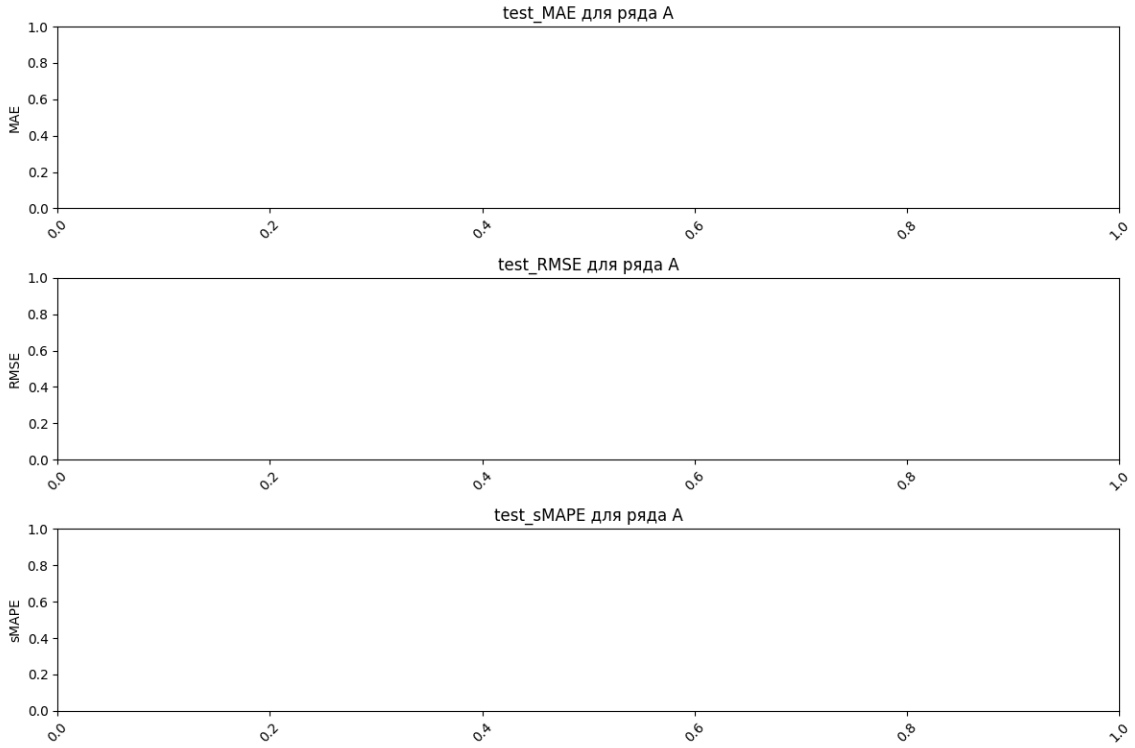
Анализ ряда A

Лучшая модель:

0
A
GRU (16u, 24w, 150e), 16b)
11.704305171966553
0.3123993601078594
0.33180706786782527
12.217951856981255
0.5437099530568105
0.5643125003292365
20.320353423394945
False

Детали модели:

Лучшая модель для ряда A: GRU (16u, 24w, 150e), 16b)
Тестовые метрики:
MAE: 0.5437
RMSE: 0.5643
sMAPE: 20.3204%
Время обучения: 11.70 сек


test_MAE для ряда A
MAE
test_RMSE для ряда A
RMSE
test_sMAPE для ряда A
sMAPE
0.0
0.2
0.4
0.6
0.8
1.0

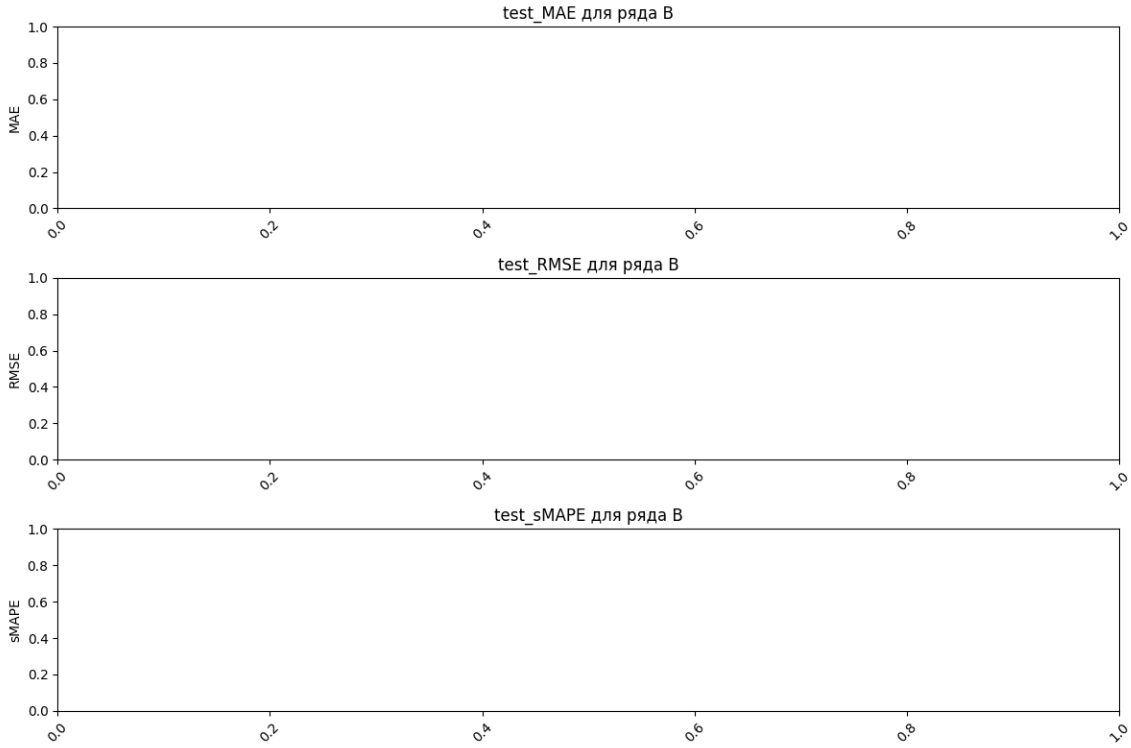
Анализ ряда B

Лучшая модель:

166
B
GRU (48u, 18w, 100e, 64b)
12.98051929473877
3.7713024616241477
4.374420919878364
3.7587790715472913
3.2698754975289996
4.020709377672101
3.2636380785920793
False

Детали модели:

Лучшая модель для ряда B: GRU (48u, 18w, 100e, 64b)
Тестовые метрики:
MAE: 3.2699
RMSE: 4.0207
sMAPE: 3.2636%
Время обучения: 12.98 сек


test_MAE для ряда B
MAE
test_RMSE для ряда B
RMSE
test_sMAPE для ряда B
sMAPE
0.0
0.2
0.4
0.6
0.8
1.0

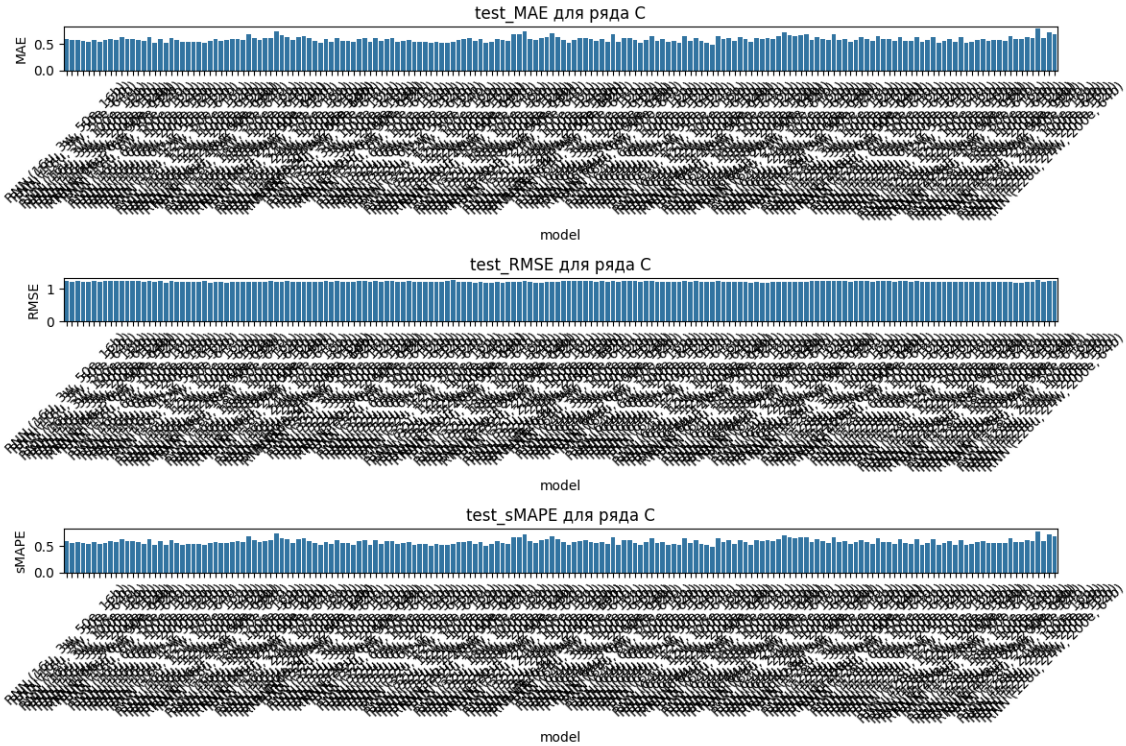
Анализ ряда C

Лучшая модель:

117
C
RNN (64u, 18w, 50e, 16b)
9.345092535018921
0.32595906850454026
0.3685242146031533
0.32479682360889506
0.4946384229158086
1.1982175344107455
0.4859517572687109
False

Детали модели:

Лучшая модель для ряда C: RNN (64u, 18w, 50e, 16b)
Тестовые метрики:
MAE: 0.4946
RMSE: 1.1982
sMAPE: 0.4860%
Время обучения: 9.35 сек


test_MAE для ряда C
MAE
0.5
0.0
model
test_RMSE для ряда C
RMSE
1
0
model
test_sMAPE для ряда C
sMAPE
0.5
0.0
model